Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **AI/ML systems**, **computer-vision workflows**, full-stack automation, SQL/NoSQL data platforms, and tested integrations for scientific, operational, and real-time applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Built Azure-backed AI education platform with normalized SQL schemas, row-level isolation, and indexed dashboards, reducing latency from 1.2 seconds to 350 milliseconds for 500+ campus users across releases.
- Reworked FastAPI background jobs with explicit states, failure isolation, and SQL-backed status tracking, processing 1K+ AI jobs while tripling throughput under concurrent instructor workloads, tickets, and reviews.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Designed concurrent file-ingestion services with ThreadPoolExecutor, OCR, and multimodal AI review, reducing feedback turnaround 60% while keeping batch processing testable, observable, and responsive for instructors and QA.
- Implemented React and Flask grading workflows for uploads, rubrics, history, and model selection, supporting 150+ instructors and 10K+ daily requests through production REST APIs, dashboards, and fixes.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built a Decision Transformer for autonomous-driving offline RL, improving planning 18% in simulation while processing 100K+ trajectory logs through reproducible Python evaluation pipelines for systems research experiments.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% across simulated baselines for repeatable performance analysis workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Troubleshot large-scale API integrations using OOP patterns, SQL workflows, AWS Glue, and S3, improving response times 30% while reducing technical debt 20% across release cycles and defects.
- Engineered cross-platform microservices with Jenkins CI/CD, VM automation, and 20+ PyTest, Mockito, unit, and integration suites while tracking defects through Agile sprint delivery and release QA reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Developed React/Node resource-management workflows with REST APIs, WebSockets, and modular services, synchronizing 2M+ records across locations with 33% fewer conflicts during staged rollout cycles and review tickets.
- Implemented CRM modules for authentication, resources, billing, leads, permissions, and notifications, using Jenkins CI/CD and Docker staging to cut deployment time 50% safely across releases and fixes.

Skills

Languages & OOP: Python, Java, C++, TypeScript, JavaScript, SQL, OOP, Design Patterns, Modular Design, Data Structures, Algorithms
AI/ML & Vision: OpenCV, TensorFlow, PyTorch, Computer Vision, Image Processing, Machine Learning, OCR, Transformers, Model Evaluation, Sensor Data
Full-Stack & IoT: React, Node.js, Flask, FastAPI, REST APIs, MQTT, WebSockets, Responsive UI, Real-Time Monitoring, Device Control
Data & Delivery: PostgreSQL, SQL Server, Azure SQL, MongoDB, NoSQL, Time-Series Data, AWS, Docker, CI/CD, Jenkins, Git, Unit Testing, Debugging

Projects

LabVision Control Console | *Python, OpenCV, TensorFlow, MQTT, React, SQL, Docker*
- Designed a laboratory automation concept with OpenCV image checks, TensorFlow-style classification, MQTT sensor events, React monitoring screens, SQL acquisition tables, Docker packaging, unit tests, and SOP-ready documentation.

PiXMod Live | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*
- Built a responsive Next.js image platform with CLIP search, SAM segmentation, Stable Diffusion editing, MongoDB storage, and Stripe billing, processing 1K+ edited visual assets across workflows online.

CaptionCraft Live | *Transformers, PyTorch, Flask, Streamlit, Gradio*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom transformer captioning models on 10K images and 100K captions, improving caption accuracy 15% while building reliable inference APIs for image workflows.